Physics 468/668: Statistical Physics **Spring 2019**

Lecture 9: Energy fluctuation and equipartition theorem

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

9.1 Energy fluctuation

We said that in the canonical ensemble, a system could take energy anywhere between zero and infinity. However, we don't know how it works in reality.

To answer this question, let's examine the range based on the expression of the mean energy.

$$U =< E >= \frac{\sum_r E_r \exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} \tag{9.1}$$

Differentiating it with respect to β, we get,

$$\frac{\partial U}{\partial \beta} = -\frac{\sum_r E_r^2 \exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} + \frac{\left[\sum_r E_r \exp(-\beta E_r)\right]^2}{\left[\sum_r \exp(-\beta E_r)\right]^2} = - < E^2 > + < E >^2 \tag{9.2}$$

from which it follows,

$$< E^2 > - < E >^2 = -\frac{\partial U}{\partial \beta} = kT^2\left(\frac{\partial U}{\partial T}\right) = kT^2 C_V \tag{9.3}$$

Therefore, the deviation is

$$\frac{\sqrt{< E^2 > - < E >^2}}{< E >} = \frac{\sqrt{kT^2 C_V}}{U} \tag{9.4}$$

Which is $O(N^{-1/2})$. Therefore, the fluctuation would become negligible when N is large.

If we consider the manner in which energy is distributed among the various members of the canonical ensemble. We treat E as a continuous variable,

$$P(E)dE \approx \exp(-\beta E) g(E) dE \tag{9.5}$$

The probability function is governed by two terms

1. the Boltzmann factor, which monotonically decreases with E
2. the density of states, which monotonically increases with E

Thereby, it must reach some extreme E^*

$$\left(\frac{\partial(\exp(-\beta E)g(E))}{\partial E}\right)|_{E=E^*} = 0 \tag{9.6}$$

That is

$$\frac{\partial \ln g(E)}{\partial E}|_{E=E^*} = \beta \tag{9.7}$$

Given that

$$S = k \ln g \tag{9.8}$$

and

$$\left(\frac{\partial S(E)}{\partial E}\right)_{E=U} = \frac{1}{T} = k\beta \tag{9.9}$$

$$E^* = U \tag{9.10}$$

This implies that the most likelihood value of energy is identical to the mean value, regardless of the physical nature of the given system.

We now expand the logarithm of $P(E)$ around E^*

$$\begin{aligned} \ln[\exp(-\beta E)g(E)] &= \left(-\beta U + \frac{S}{k}\right) + \frac{1}{2}\frac{\partial^2}{\partial E^2}\ln\left[\exp(-\beta E)g(E)\right]|_{E=U}(E-U)^2 + ... \\ &= -\beta(U - TS) - \frac{1}{2kT^2C_V}(E-U)^2 + ... \end{aligned} \tag{9.11}$$

From which we obtain,

$$P(E) \propto \exp(-\beta E)g(E) \approx \exp(-\beta(U-TS))\exp\left[-\frac{(E-U)^2}{2kT^2C_V}\right] \tag{9.12}$$

This is a Gaussian distribution, with mean value U and dispersion $\sqrt{kT^2C_V}$.

Take the example of ideal gas,

9.2 Equipartition Theorem

To derive these theorems, we determine the expectation value of the quantity $x_i(\partial H/\partial x_j)$, where $H(q,p)$ is the Hamiltonian of the system while x_i and x_j are any two of the $6N$ coordinates (q,p).

In the canonical ensemble,

$$\left\langle x_i \frac{\partial H}{\partial x_j}\right\rangle = \frac{\int (x_i \frac{\partial H}{\partial x_j})\exp(-\beta H)d\omega}{\int \exp(-\beta H)d\omega} \tag{9.13}$$

Integrating over x_j by parts, the numerator becomes,

$$\int\left[-\frac{1}{\beta}x_i e^{-\beta H}|_{(x_j)_1}^{(x_j)_2} + \frac{1}{\beta}\int\left(\frac{\partial x_i}{\partial x_j}e^{-\beta H}dx_j\right)\right]d\omega(j) \tag{9.14}$$

Here $(x_j)_1$ and $(x_j)_2$ are the extreme values of the coordinate x_j. The integrated part here vanishes because whenever any of the coordinates takes an extreme value the Hamiltonian of the system becomes infinite. In the integral that remains, the factor $\partial x_i/\partial x_j$, being equal to δ_{ij}, comes out of the integral sign and we are left with

$$\frac{1}{\beta}\delta_{ij}\int e^{-\beta H}d\omega \tag{9.15}$$

Substituting this into (1), we get,

$$\left\langle x_i\frac{\partial H}{\partial x_j}\right\rangle = \delta_{ij}kT \tag{9.16}$$

Which is independent of the precise form of the function H.

In the special case $x_j = x_i = p_i$, the above equation becomes,

$$\left\langle p_i\frac{\partial H}{\partial p_i}\right\rangle =< p_i\dot{q}_i >= kT \tag{9.17}$$

In the special case $x_j = x_i = q_i$, the above equation becomes,

$$\left\langle q_i\frac{\partial H}{\partial q_i}\right\rangle = - < q_i\dot{p}_i >= kT \tag{9.18}$$

Adding over all i, we obtain

$$\left\langle \sum_i p_i\frac{\partial H}{\partial p_i}\right\rangle =< \sum_i p_i\dot{q}_i >= 3NkT \tag{9.19}$$

$$\left\langle \sum_i q_i\frac{\partial H}{\partial q_i}\right\rangle = - < \sum_i q_i\dot{p}_i >= 3NkT \tag{9.20}$$

Now, if H is composed of quadratic functions,

$$H = \sum_j A_jP_j^2 + \sum_j B_jQ_j^2 \tag{9.21}$$

where P_j and Q_j are canonically conjugate, We clearly have

$$\sum_j(P_j\frac{\partial H}{\partial P_j} + Q_j\frac{\partial H}{\partial Q_j}) = 2H \tag{9.22}$$

accordingly,

$$< H >= \frac{1}{2}fkT \tag{9.23}$$

Where f is the number of non-vanishing coefficients, namely, the number of DOF. It is valid only when the relevant of DOF can be freely excited.

9.3 Virial Theorem

The sum of products of the coordinates of the particles and the respective forces acting on them is referred to as the *virial* of the system and is denoted by the $\mathcal{V}$

$$\mathcal{V} = -3NkT \tag{9.24}$$

The relationship between the $\mathcal{V}$ and other quantities is best understood by looking at the cases of gas. In this case, the forces are the ones arising from the walls; these forces can be designated by an external pressure P that acts on the system by virtue of the fact that it is bounded by the walls. Consequently, we have a force $-PdS$ associated with an element of dS, the negative sign appears because the force is inward, while dS is outward. The Virial of the gas is given by

$$\mathcal{V}_0 = (\sum_i q_i F_i)_0 = -P \int r \cdot dS \tag{9.25}$$

where r is the position vector. By the divergence theorem,

$$\mathcal{V}_0 = -P \int (\nabla \cdot r) dV = -3PV \tag{9.26}$$

Of course, we can find

$$PV = NkT \tag{9.27}$$

$$\mathcal{V} = -2K \tag{9.28}$$

where K is the kinetic energy.

The Virial equation of states is expressed as follows

$$\frac{P}{NkT} = 1 + \frac{1}{NdkT} \left\langle \sum_{i<j} F(r_{ij}) \cdot r_{ij} \right\rangle = 1 - \frac{1}{NdkT} \left\langle \sum_{i<j} \frac{\partial u(r_{ij})}{\partial r_{ij}} r_{ij} \right\rangle \tag{9.29}$$

9.4 Homework

1. 3.5
2. 3.8
3. 3.9
4. 3.13
5. 3.15